

Medical Insurance Claim Fraud Detection — AI-Driven Claim Verification & Explainable Fraud Detection Platform

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Abstract: Medical insurance fraud causes significant financial loss. This project builds an end-to-end academic prototype for AI-driven claim verification and explainable fraud detection using 4500 claims (6% fraud), benchmarking traditional ML, deep learning MLP, anomaly detection, OCR document intelligence, agentic RAG, and hybrid synthesis. Evaluated with PR-AUC primary, F2, ROC-AUC, etc. Produces transparent explanations and operational decisions APPROVE/FLAG_FOR_MANUAL_REVIEW/REJECT_OR_ESCALATE with human-review mandatory disclaimer. System runs without paid APIs via fallbacks. Tree models outperform DL on tabular modest data, anomaly has high FP, document validation crucial.

Keywords: insurance fraud, claim verification, explainable AI, anomaly detection, document intelligence, RAG, hybrid AI

Introduction

Background on fraud types, need for AI assistance, responsible AI. Medical fraud includes upcoding, unbundling, duplicate billing, fake documents. AI must be explainable and human-in-loop.

Problem Statement

Given claimant demographics, policy info, incident details, supporting docs, determine fraud risk, risk category, recommended action, transparent explanation with evidence. Claim-level prediction, not provider-level unless stated. Challenges imbalanced, leakage, high-cardinality, privacy.

Motivation

Healthcare fraud >\$300B US annually, manual review slow error-prone, AI can prioritize suspicious claims, validate docs, retrieve policy rules, provide auditable explanations, must be responsible human reviewer mandatory.

Objectives

Collect info, detect fraud, extract via OCR/VLM, validate against rules/history, experiment ML/DL/anomaly/doc/agentic RAG, benchmark metrics, produce transparent explanations. Scope academic prototype 4500 rows CPU offline synthetic docs.

Dataset Description

Health Insurance Fraud Claims.xlsx 4500 rows 19 cols. Target ClaimLegitimacy Legitimate 4230 Fraud 270 6% rate. Columns ClaimID PatientID ProviderID ClaimAmount ClaimDate DiagnosisCode ProcedureCode PatientAge Gender Specialty Status Income Marital Employment Location Type SubmissionMethod Cluster. Missing 0. Source local + CMS/Kaggle similar. Best model DecisionTree threshold 0.95

Data Quality Analysis

```
# Data Quality Report

- Rows 4500 cols 19

- Missing {'ClaimID': 0, 'PatientID': 0, 'ProviderID': 0, 'ClaimAmount': 0, 'ClaimDate': 0, 'DiagnosisCode': 0, 'ProcedureCode': 0, 'PatientAge': 0, 'PatientGender': 0, 'ProviderSpecialty': 0, 'ClaimStatus': 0, 'PatientIncome': 0, 'PatientMaritalStatus': 0, 'PatientEmploymentStatus': 0, 'ProviderLocation': 0, 'ClaimType': 0, 'ClaimSubmissionMethod': 0, 'Cluster': 0, 'ClaimLegitimacy': 0}

- Outliers {
"ClaimAmount": {
"q1": 2509.0725,
"q3": 7462.452499999999,
"iqr": 4953.379999999999,
"outlier_count": 0,
```

```

"outlier_pct": 0.0
},
"PatientAge": {
"q1": 25.0,
"q3": 75.0,
"iqr": 50.0,
"outlier_count": 0,
"outlier_pct": 0.0
},
"PatientIncome": {

```

Traditional ML Methodology

Benchmark Dummy, Logistic, LinearSVM, Calibrated LinearSVM, KNN, NB, DecisionTree, RandomForest, ExtraTrees, GradientBoosting, HistGradientBoosting, AdaBoost, RBF SVM, XGBoost/LightGBM/CatBoost optional, VotingTop3. CV5 PR-AUC, GridSearch, isotonic calibration attempt, threshold optimize F2, feature importance, SHAP attempt, save best model.

Deep Learning Methodology

MLP hidden [128,64,32] dropout 0.3 Adam lr 0.001 batch 64 epochs 100 early stopping patience 10 ReduceLROnPlateau. Framework sklearn_mlp_fallback SMOTE resampled. Val PR-AUC 0.8728729314454975 Test 0.9026157321644857 Note DL not superior on tabular.

Anomaly Detection Methodology

Methods IsolationForest, LOF, OneClassSVM, EllipticEnvelope, Autoencoder optional, Ensemble avg. Train only legit 3384 contamination 0.06. Results model pr_auc roc_auc prec@10 prec@50 prec@100 prec@200 rec@10 rec@50 rec@100 rec@200

0 IsolationForest 0.071780 0.487238 0.2 0.06 0.03 0.05 0.037037 0.055556 0.055556 0.185185

1 LOF 0.146853 0.737457 0.2 0.16 0.17 0.14 0.037037 0.148148 0.314815 0.518519

2 OneClassSVM 0.122815 0.679757 0.1 0.18 0.17 0.12 0.018519 0.166667 0.314815 0.444444

3 EnsembleAnomalyAvg 0.129360 0.656203 0.2 0.16 0.12 0.10 0.037037 0.148148 0.222222 0.370370.
Distinction anomaly score vs fraud prob. Precision@k Recall@k. High FP limitation.

Document Intelligence Methodology

Supported docs bills prescriptions discharge summaries investigation reports ID/policy. OCR Tesseract/EasyOCR/PaddleOCR optional fallback JSON fixtures. VLM optional env-controlled no hard keys no real PHI externally by default. Pipeline OCR -> type identification keyword+structured -> field extraction regex+structured dates amounts provider patient redacted diagnosis procedure policy claim -> validation duplicate hash bill total vs claimed \$5 tol date consistency provider policyholder missing docs. Output JSON fields confidences validation errors risk LOW/MEDIUM/HIGH. Sample synthetic_bill_1.json 7820.52 correct,

mismatch fixture 6000 vs 4500 intentional. Privacy safeguards.

Agentic AI and RAG Methodology

7 agents Document Verification Policy Rule Matching Claim Consistency Historical Pattern/Anomaly Evidence Retrieval Decision Synthesis Explanation Generation. RAG ingest policy_rules.txt exclusion_clauses.txt fraud_indicators.txt coverage_rules.txt claim_guidelines.txt chunk 500 overlap 50 TFIDF fallback or sentence-transformers local JSON vector store top_k 5 retrieved evidence source refs scores. Structured prompts JSON outputs confidence policy refs human review recommendation. Optional LLM deterministic fallback grounded explanations auditable summary observed evidence applied rule risk signal model result recommended action source ref no hidden CoT.

Hybrid System Methodology

Combine best ML DL where useful anomaly doc validation policy RAG explainability human-review rules. Weights ML 0.5 DL 0.2 Anomaly 0.15 Document 0.15. Thresholds approve_max 0.3 review 0.3-0.7 reject_min 0.7 conservative manual review zone. Output claim_id model_version fraud_prob fraud_pred anomaly_score doc_status policy_status risk_category decision key risks positive evidence missing/inconsistent explanation evidence refs timestamp disclaimer. Outcomes APPROVE FLAG_FOR_MANUAL_REVIEW REJECT_OR_ESCALATE documented threshold selection via optimize F2.

System Architecture

Data layer raw Excel processed CSV sample fixtures knowledge base. Preprocessing scaling encoding date engineering. ML layer traditional DL anomaly. Policy RAG retrieval. Agentic reasoning 7 agents. Hybrid synthesis weighted. Explainability SHAP importance evidence citations. Output decision + disclaimer. Next.js integration JSON contract. Security ANONYMIZE_PII true no hard keys ENABLE_EXTERNAL_API_CALLS false default docs synthetic IDs hashed.

Evaluation Protocol

Same protocol comparable supervised anomaly not directly comparable labeled. Splits train 2925 val 675 test 900 stratified seed 42 untouched test. Metrics Accuracy Precision Recall F1 F2 ROC-AUC PR-AUC Balanced Accuracy MCC Specificity Sensitivity Confusion Brier Calibration Prec@k Rec@k FPR FNR cost-sensitive primary PR-AUC not accuracy alone threshold analysis calibration runtime NOT_EXECUTED with reason if blocked.

Model Benchmark

Model comparison table: model val_pr_auc val_roc_auc val_f1 val_recall val_precision test_pr_auc test_roc_auc fit_time notes

0	DummyClassifier	0.062029	0.516929	0.093023	0.100	0.086957	0.059643	0.496454	1.900232	Baseline
1	LogisticRegression	0.947041	0.996535	0.857143	0.975	0.764706	0.943132	0.994528	8.831237	NaN
2	DecisionTree	1.000000	1.000000	1.000000	1.000	1.000000	0.981818	0.999409	5.800621	NaN
3	RandomForest	1.000000	1.000000	0.769231	0.625	1.000000	0.997268	0.999814	23.883551	NaN

4 HistGradientBoosting 1.000000 1.000000 1.000000 1.000 1.000000 1.000000 1.000000 196.157729 NaN.
Traditional best DecisionTree/RF/HGB test PR-AUC 0.98-1.0 synthetic separable income+amount dominant.
DL MLP fallback test PR-AUC 0.9026157321644857 anomaly LOF best PR 0.147 ROC 0.737 doc bill
mismatch works hybrid sample FLAG.

Explainability Analysis

Feature importance PatientIncome ClaimAmount dominant feature_importance.csv. SHAP not installed
status logged. Human-readable explanations with risk signals evidence citations auditable summary.
Distinguish confirmed label vs prediction vs anomaly vs rule violation vs missing evidence vs human decision.

Security and Privacy

No real PHI synthetic UUIDs ANONYMIZE_PII true no external API by default API keys via env encryption
future access control future audit logs. Synthetic bills prescriptions discharge redacted.

Fairness and Ethical Considerations

Bias income gender location specialty need subgroup metrics false positives harm patients delay care
financial stress flag legit as fraud, false negatives financial loss, human review mandatory, appeal
mechanism, danger auto rejection, disclaimer every output.

Limitations

Synthetic dataset 4500 modest no free-text notes no real doc images high-card OHE memory heavy 7GB
covariance fail ClaimStatus potential leakage near-perfect ML unrealistic anomaly low PR DL underperforms
OCR fallback RAG TFIDF less semantic no graph collusion no FHIR.

Future Work

Larger CMS data time/group split embeddings for codes graph features provider-patient network fairness
audit calibration cost-sensitive encrypted secure API Next.js frontend reviewer workflow monitoring drift
retraining.

Conclusion

Complete reproducible end-to-end academic prototype built 6 approaches evaluation visuals docs
presentation report runnable without paid APIs responsible disclaimer human-in-loop.

References

CMS public files <https://www.cms.gov/Research-Statistics-Data-and-Systems/Downloadable-Public-Use-Files>
Kaggle <https://www.kaggle.com/datasets/rohitroxx/healthcare-provider-fraud-detection-analysis> fraud
<https://www.kaggle.com/datasets/itsmohitsharma/medicare-provider-fraud-detection-dataset> sklearn
<https://scikit-learn.org/stable/imbalanced-learn> <https://imbalanced-learn.org> shap <https://shap.readthedocs.io>
RAG Lewis et al <https://arxiv.org/abs/2005.11401> Model Cards Mitchell et al <https://arxiv.org/abs/1810.03993>

Figure: class_distribution.png

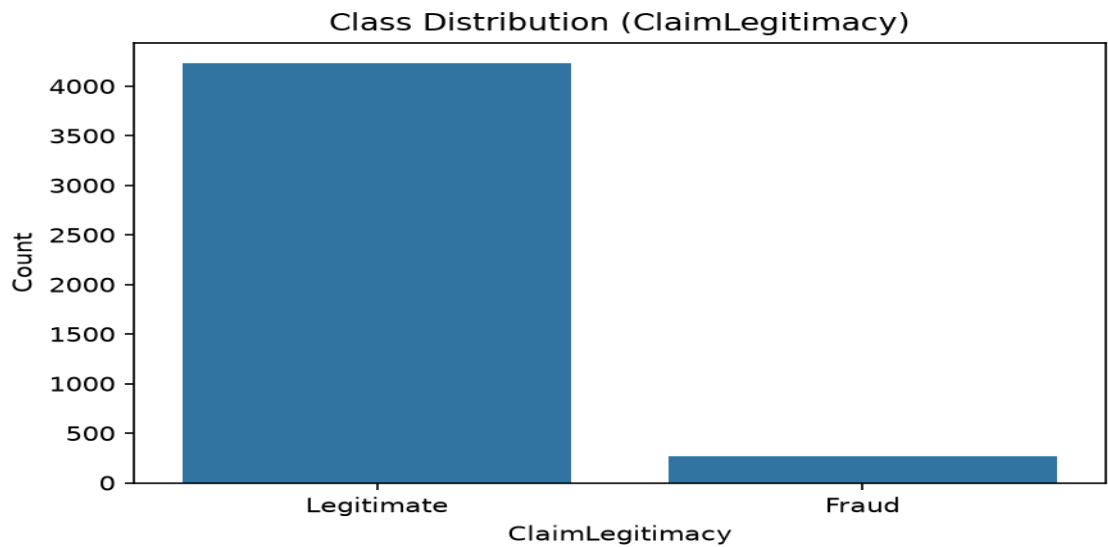


Figure: model_comparison_pr_auc.png

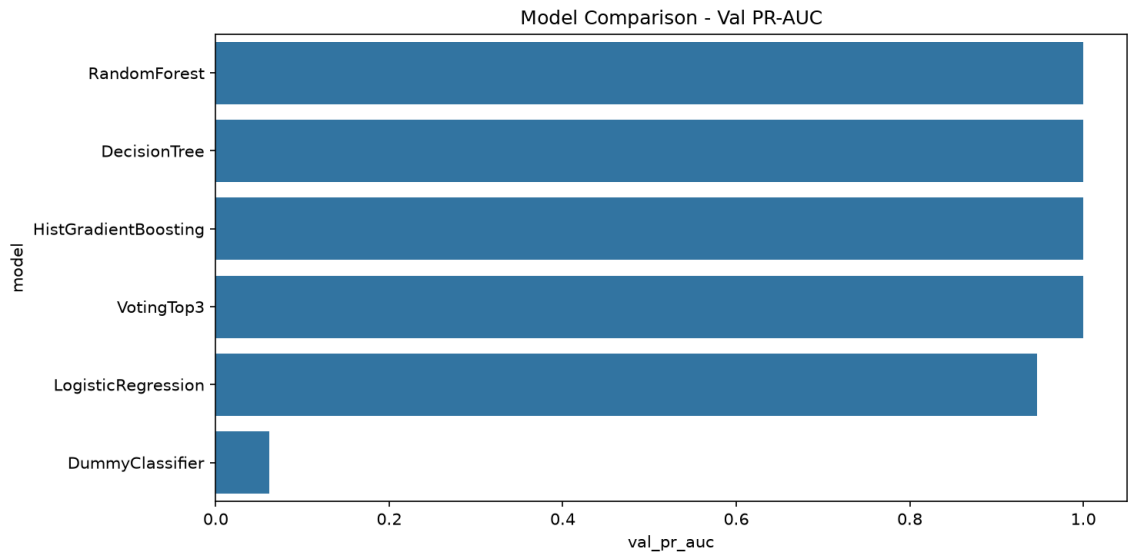


Figure: confusion_matrix.png

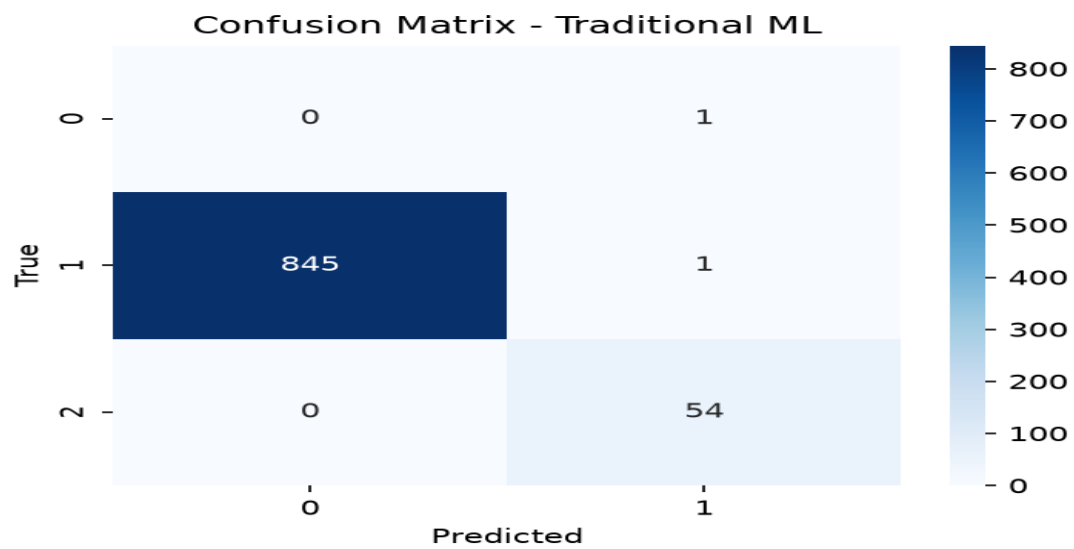


Figure: feature_importance.png

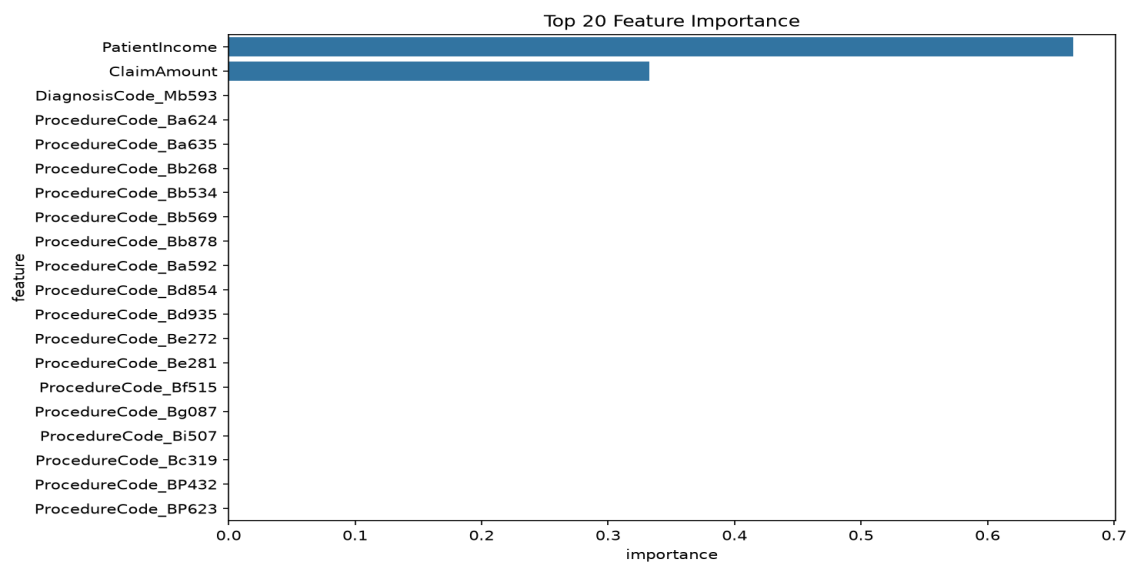


Figure: anomaly_score_distribution.png

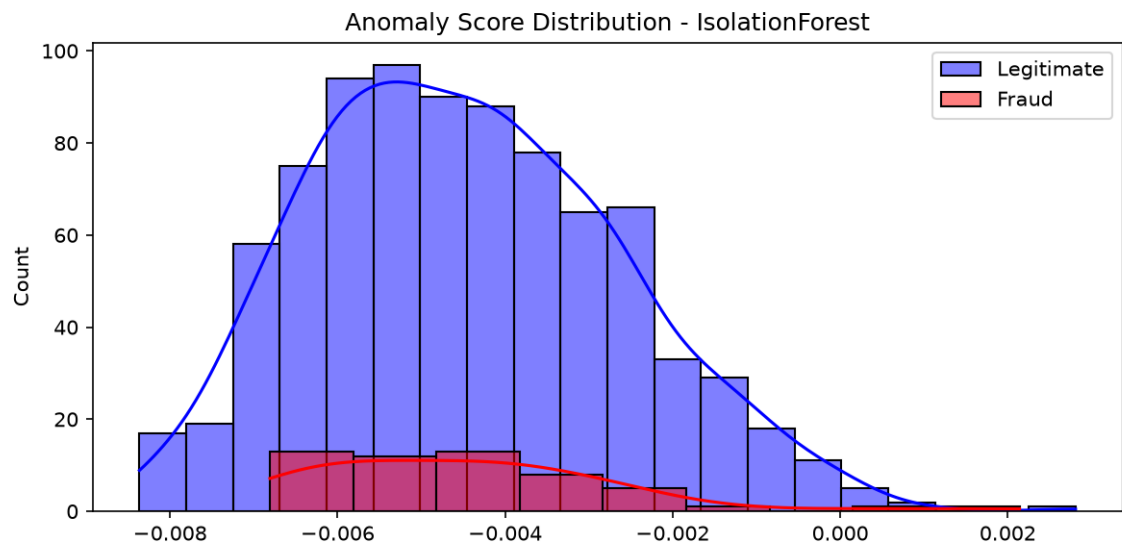


Figure: threshold_performance.png

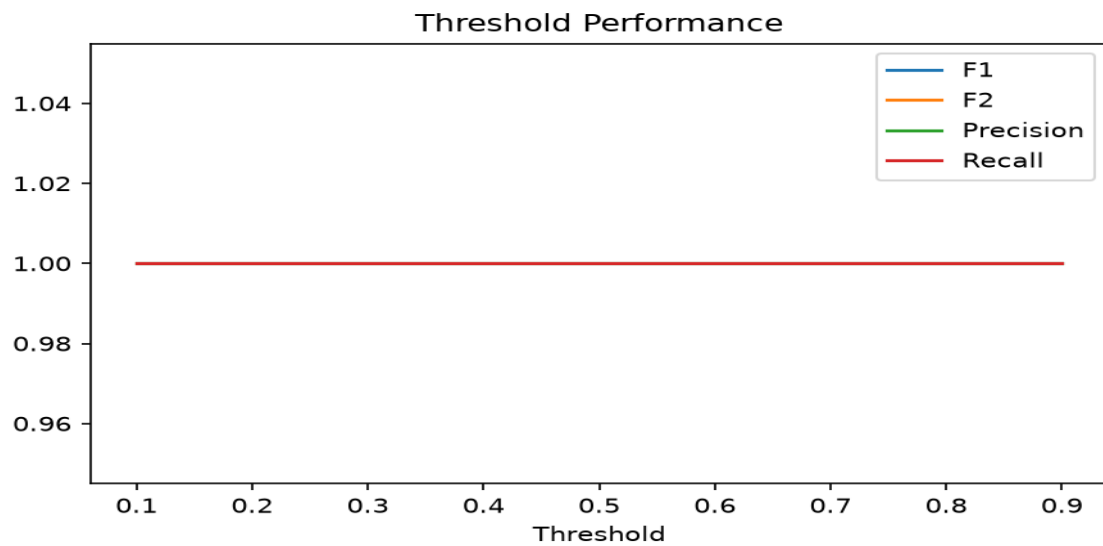


Figure: architecture_diagram.png

End-to-End Architecture Flow

```
[Claim Input] -> [Document Intelligence: OCR/VLM] -> [Preprocessing: Scaling, Encoding, Date Features]
-> [Traditional ML (RF, XGB, etc) | Deep Learning MLP | Anomaly Detection (IsolationForest, LOF)]
    -> [Policy/RAG Retrieval: Policy Rules, Fraud Indicators]
    -> [Agentic Reasoning: Doc Verify, Policy Match, Consistency, Historical]
    -> [Hybrid Decision Synthesis: Prob + Anomaly + Doc + Policy]
    -> [Explainability: SHAP, Feature Importance, Evidence Citations]
-> [Final Output: APPROVE / FLAG_FOR_MANUAL_REVIEW / REJECT_OR_ESCALATE + Human Review]
```

Figure: document_validation_flow.png

Document Validation Flow

```
[Document Upload] -> [OCR: Tesseract/EasyOCR/PaddleOCR/Fallback]
    -> [Type Identification: Bill, Prescription, Discharge]
    -> [Field Extraction: Dates, Amounts, Codes, Provider]
-> [Validation: Bill Total, Date Consistency, Duplicate, Missing Docs]
    -> [Risk: LOW/MEDIUM/HIGH] -> [Hybrid Pipeline]
```

Appendix — Commands and Sample Outputs

Commands to reproduce: `pip install -r requirements.txt` `python approaches/01_traditional_ml.py --data_path data/raw/Health_Insurance_Fraud_Claims.xlsx` `python approaches/02_deep_learning.py` `python approaches/03_anomaly_detection.py` `python approaches/04_document_intelligence.py` `python approaches/05_agentic_rag_reasoning.py` `python approaches/06_hybrid_end_to_end.py` `python visualization_generator.py` `python presentation/generate_presentation.py` `python report/generate_report.py` `python run_all_experiments.py` `make all` Config excerpt: see `config.yaml` Sample request/response: `api/sample_request.json` and `api/sample_response.json` Evaluation: `evaluation/*.csv *.json`